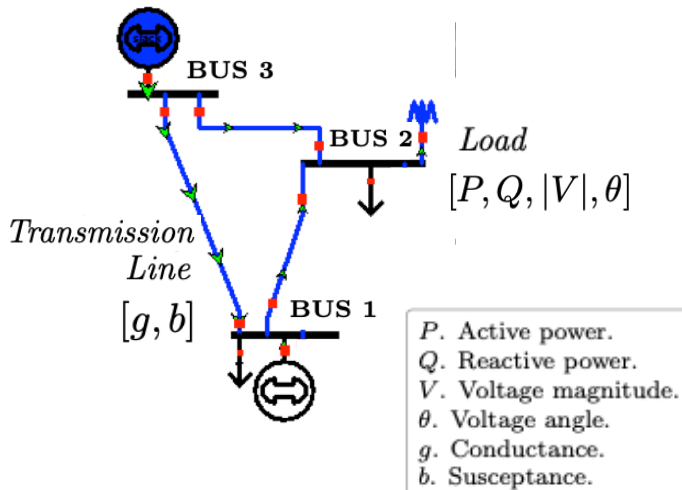
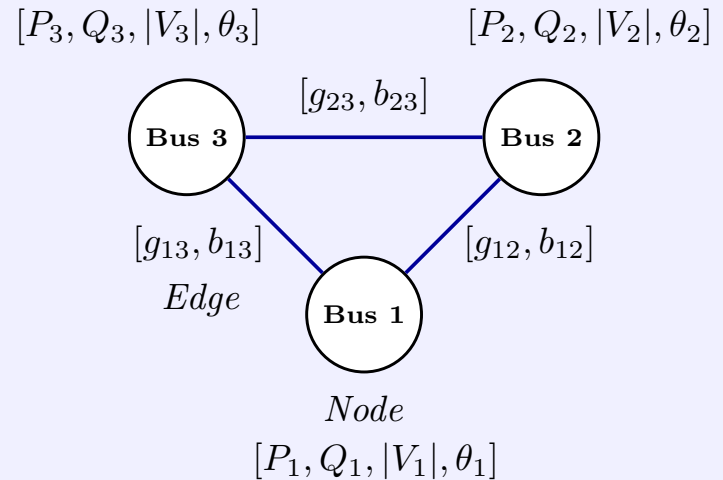


Power System Diagram



*Graph
Construction*

Graph Abstraction



*Feature
Assignment*

Matrix Representation

$$A = \begin{matrix} & v & u_1 & u_2 \\ \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} & \begin{matrix} v \\ u_1 \\ u_2 \end{matrix} \end{matrix}$$

$$X = \begin{bmatrix} h_{\{v,1\}} & \cdots & h_{\{v,d_f\}} \\ h_{\{u_1,1\}} & \cdots & h_{\{u_1,d_f\}} \\ h_{\{u_2,1\}} & \cdots & h_{\{u_2,d_f\}} \end{bmatrix}$$

$$E = \begin{bmatrix} e_{\{u_1 v,1\}} & \cdots & e_{\{u_1 v,d_e\}} \\ e_{\{u_2 v,1\}} & \cdots & e_{\{u_2 v,d_e\}} \end{bmatrix}$$

*Matrix
Encoding*

Message Passing Mechanism

